C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 2.01

Neural Layer Matrices & Activations

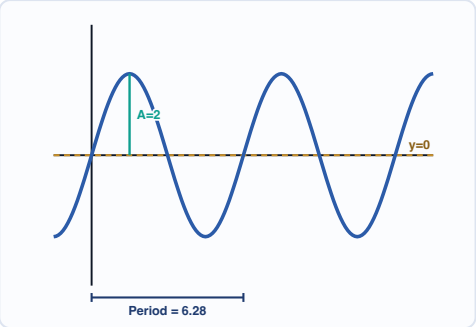
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Neural Layer Matrices & Activations Practice Exercises

1 MCQ

A fully-connected dense layer maps an input $\mathbf{x} \in \mathrm{R}^{50}$ to $\mathbf{y} \in \mathrm{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?



A 1275

B 1250

C 75

D 1325

2 MCQ

A fully-connected dense layer maps an input $\mathbf{x} \in \mathrm{R}^{60}$ to $\mathbf{y} \in \mathrm{R}^{30}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

A 1830

B 1800

C 90

D 1890

3 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -4$ and $z_2 = 6$.

A $f(z_1) = 4,\ f(z_2) = 6$

B $f(z_1) = -4,\ f(z_2) = 6$

C $f(z_1) = 0,\ f(z_2) = 0$

D $f(z_1) = 0,\ f(z_2) = 6$

4 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -6$ and $z_2 = 9$.

A $f(z_1) = 6,\ f(z_2) = 9$

B $f(z_1) = -6,\ f(z_2) = 9$

C $f(z_1) = 0,\ f(z_2) = 0$

D $f(z_1) = 0,\ f(z_2) = 9$

5 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -10$ and $z_2 = 15$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -10,\ f(z_2) = 15$

C $f(z_1) = 0,\ f(z_2) = 15$

D $f(z_1) = 10,\ f(z_2) = 15$

6 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -12$ and $z_2 = 18$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -12,\ f(z_2) = 18$

C $f(z_1) = 0,\ f(z_2) = 18$

D $f(z_1) = 12,\ f(z_2) = 18$

7 MCQ

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.20$, what is the value of the derivative $\sigma'(z_0)$?

A 0.1600

B 0.4000

C 0.8000

D 0.0400

8 MCQ

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

A 0.2400

B 0.8000

C 0.6000

D 0.1600

9 MCQ

Determine the numerical value of the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -6$ and $z_2 = 9$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = 0,\ f(z_2) = 9$

C $f(z_1) = 6,\ f(z_2) = 9$

D $f(z_1) = -6,\ f(z_2) = 9$

10 MCQ

Calculate the simplified result of the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -10$ and $z_2 = 15$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -10,\ f(z_2) = 15$

C $f(z_1) = 10,\ f(z_2) = 15$

D $f(z_1) = 0,\ f(z_2) = 15$

11 MCQ

Find the exact value of the expression the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -12$ and $z_2 = 18$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -12,\ f(z_2) = 18$

C $f(z_1) = 0,\ f(z_2) = 18$

D $f(z_1) = 12,\ f(z_2) = 18$

12 MCQ

In this problem, the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

A 0.1600

B 0.8000

C 0.2400

D 0.6000

13 MCQ

Observe that the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.60$, what is the value of the derivative $\sigma'(z_0)$?

A 0.4000

B 0.3600

C 1.2000

D 0.2400

14 MCQ

Suppose the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.80$, what is the value of the derivative $\sigma'(z_0)$?

A 1.6000

B 0.2000

C 0.1600

D 0.6400

15 MCQ

Notice that the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.00$, what is the value of the derivative $\sigma'(z_0)$?

A 2.0000

B 1.0000

C 13

D 0.0000

16 MCQ

Consider the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

A -0.2000

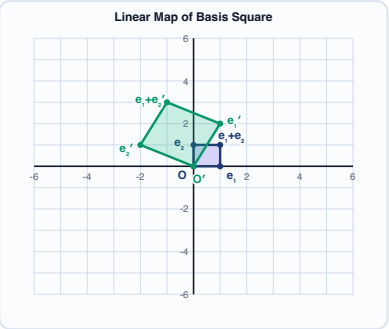
B 1.4400

C -0.2400

D 2.4000

17 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 9 \\ 1 \end{bmatrix}$

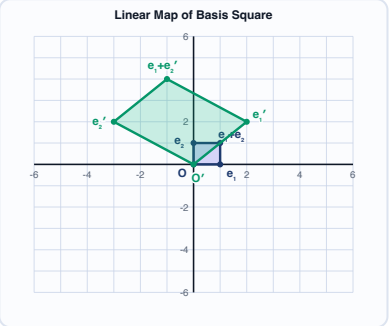
B $\begin{bmatrix} 0 \\ 9 \end{bmatrix}$

C $\begin{bmatrix} -5 \\ 9 \end{bmatrix}$

D $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$

18 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 2 & -3 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 2 \\ 12 \end{bmatrix}$

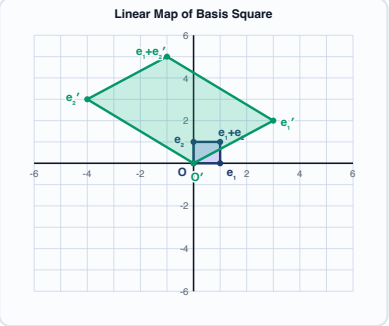
B $\begin{bmatrix} 12 \\ 1 \end{bmatrix}$

C $\begin{bmatrix} -2 \\ 12 \end{bmatrix}$

D $\begin{bmatrix} 0 \\ 12 \end{bmatrix}$

19 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 3 & -4 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 15 \\ 6 \end{bmatrix}$

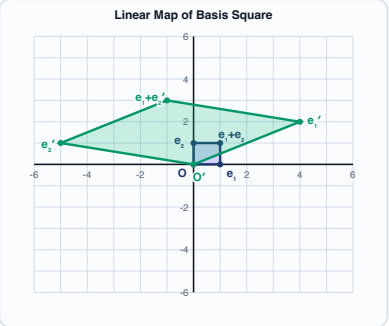
B $\begin{bmatrix} 3 \\ 15 \end{bmatrix}$

C $\begin{bmatrix} 7 \\ 15 \end{bmatrix}$

D $\begin{bmatrix} 5 \\ 15 \end{bmatrix}$

20 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 4 & -5 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 10 \\ 15 \end{bmatrix}$

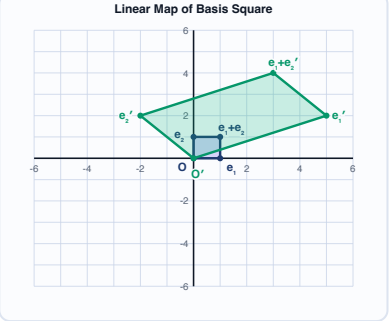
B $\begin{bmatrix} 12 \\ 15 \end{bmatrix}$

C $\begin{bmatrix} 15 \\ 13 \end{bmatrix}$

D $\begin{bmatrix} 14 \\ 15 \end{bmatrix}$

21 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 5 & -2 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 25 \\ 18 \end{bmatrix}$

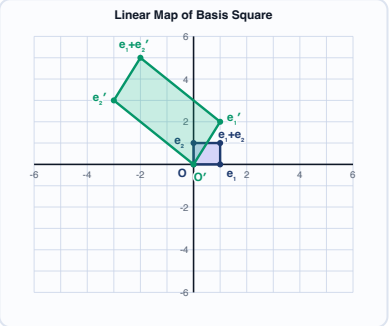
B $\begin{bmatrix} 23 \\ 18 \end{bmatrix}$

C $\begin{bmatrix} 18 \\ 26 \end{bmatrix}$

D $\begin{bmatrix} 27 \\ 18 \end{bmatrix}$

22 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 1 & -3 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} -1 \\ 21 \end{bmatrix}$

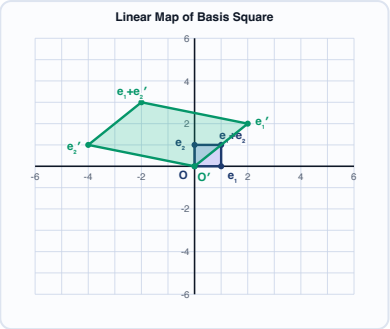
B $\begin{bmatrix} 3 \\ 21 \end{bmatrix}$

C $\begin{bmatrix} 1 \\ 21 \end{bmatrix}$

D $\begin{bmatrix} 21 \\ 2 \end{bmatrix}$

23 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 2 & -4 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} -9 \\ 10 \end{bmatrix}$

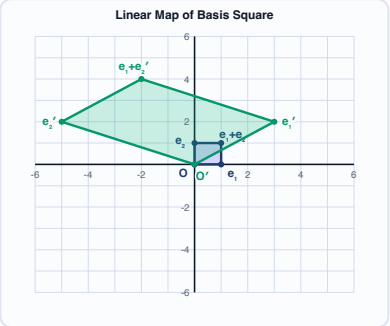
B $\begin{bmatrix} 9 \\ 10 \end{bmatrix}$

C $\begin{bmatrix} 10 \\ 1 \end{bmatrix}$

D $\begin{bmatrix} 0 \\ 10 \end{bmatrix}$

24 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 3 & -5 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 14 \\ 1 \end{bmatrix}$

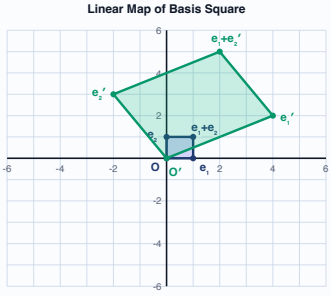
B $\begin{bmatrix} -6 \\ 14 \end{bmatrix}$

C $\begin{bmatrix} 0 \\ 14 \end{bmatrix}$

D $\begin{bmatrix} 6 \\ 14 \end{bmatrix}$

25 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 4 & -2 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 11 \\ 18 \end{bmatrix}$

B $\begin{bmatrix} 9 \\ 18 \end{bmatrix}$

C $\begin{bmatrix} 7 \\ 18 \end{bmatrix}$

D $\begin{bmatrix} 18 \\ 10 \end{bmatrix}$